

A Survey of Multi-Paradigm Approaches Toward General-Purpose Embodied Intelligence: Panoramic Overview and Evolutionary Trajectory

Jiaxin Zhuang, Tao Zhou, Rule Shi, Yunlong Liu, Qinghui Chen, Zekai Zhang, Wei Zhang, and Jinglin Zhang

Abstract—Vision-Language-Action Models (VLAs) incorporate knowledge acquired through vision-language pretraining into robotic control, providing a unified modeling paradigm for general-purpose policies conditioned on natural-language instructions. However, existing studies are distributed across distinct research threads, including action decoding, deployment efficiency, and dynamics prediction, while current taxonomies remain insufficient to reveal how these technical choices jointly affect control performance and real-world applicability. To address this limitation, this survey develops an analytical framework centered on action-generation mechanisms that connects model architectures, deployment constraints, and predictive extensions. Within this framework, we examine how different action-decoding paradigms balance semantic transfer and continuous control, and analyze the constraints imposed by inference overhead on real-world deployment. Drawing on research on model efficiency and cross-scenario generalization, we further use representative benchmark results to assess the extent to which reported performance gains remain valid in open environments. We also clarify the technical scope of Video-Action Models (VAM) and World Action Models (WAM), showing that they do not simply replace conventional VLAs; rather, they broaden the information available for action decision-making by incorporating temporal dynamics or constraints derived from predicted future states. Our analysis suggests that VLA research is gradually expanding from localized methodological improvements toward the development of system-level capabilities for real-world operation. Future progress will therefore require closer alignment between model capabilities and deployment requirements, together with stronger long-horizon decision-making and continual adaptation in open environments. On this basis, we identify the key challenges associated with real-world deployment and the extension of general-purpose capabilities, and discuss future directions from the perspectives of predictive decision-making, data feedback loops, and system-level coordination.

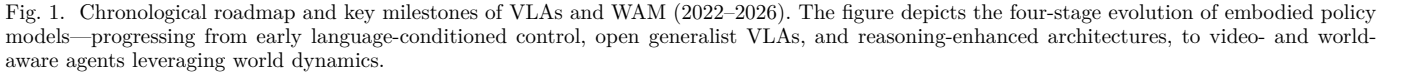
I. Introduction

In recent years, artificial intelligence research has been shifting from cognitive intelligence, centered on static perception and semantic understanding, toward embodied intelligence oriented toward interaction with the physical world. Embodied agents must not only understand natural

language instructions and perceive complex environments, but also generate executable actions and continuously adjust these actions based on environmental feedback. Vision-Language-Action Models (VLAs) provide an important technical path toward general-purpose robotic policies by unifying visual observations, language instructions, and robot control within an end-to-end framework. RT-1 [1] validated the scalability of Transformers for large-scale, multi-task robotic control; RT-2 [2] transferred knowledge from vision-language pretraining to action generation; RT-X [3] further explored joint training across tasks and robots; OpenVLA [4] promoted the open-sourcing of models and training pipelines; and π_0 [5] adopted a flow-matching mechanism to generate continuous action chunks. These approaches each have advantages in inference efficiency, semantic modeling, or continuous control, resulting in different trade-offs among control precision, generation speed, and model expressiveness.

Although VLAs have made significant progress in multi-task control and semantic generalization, their real-world deployment remains constrained by computational overhead, data dependency, environmental variation, and limitations in long-horizon execution. In terms of model efficiency, visual token pruning, quantization, knowledge distillation, and lightweight backbones can reduce computational load and storage cost, but model-level compression does not necessarily translate into high-frequency, low-latency, and stable closed-loop control in real robotic systems. In terms of scene generalization, data resources such as Open X-Embodiment [3], DROID [6], and Bridge-Data V2 [7] have expanded the coverage of tasks, objects, and robot embodiments, while benchmarks such as LIBERO [8], RoboCasa [9], and RoboTwin 2.0 [10] have introduced richer task configurations and environmental variation. However, high success rates in standard or simple scenarios do not imply that a model can reliably handle variations in camera viewpoint, object layout, lighting, background, or robot initial state; even minor perturbations outside the training distribution can still cause a marked performance drop. Meanwhile, methods such as reinforcement learning, memory mechanisms, hierarchical planning, and world models attempt to enhance a policy's interactive optimization, skill composition, and long-horizon reasoning capabilities, but still face issues such as low sample efficiency, high inference cost, error

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Beyond improving the efficiency and generalization of action prediction itself, recent research has begun to incorporate video dynamics and future world states into robotic policy learning. Early world-model research showed that an agent can learn a latent dynamic representation of the environment to predict subsequent states and perform behavioral rollouts within an internal model [15]. Building on this, Video-Action Models (VAM) leverage temporal changes, object motion, and interaction information within videos to assist action generation—for example, UVA establishes a shared video-action representation by jointly learning future observations and actions [16]; World Action Models (WAM) further emphasize the joint modeling of future world states and action sequences, enabling policies to account for the environmental changes that actions may induce [17]. VAM and WAM are not simple replacements for existing VLAs, but rather emerging directions that extend VLAs toward modeling temporal dynamics, action consequences, and future states. Existing surveys of robotic foundation models mainly focus on model types, learning paradigms, tasks, and applications [11], [18], [19]; VLA surveys focus on model compo-

Although existing surveys have advanced this field from multiple angles, there remains a lack of a unified analytical framework centered on VLAs, structured around action-generation architecture as the technical backbone, using lightweight design and scene generalization as deployment through-lines, and further connecting to the frontier paradigms of VAM/WAM. Based on this, this paper presents a systematic survey of VLAs and their frontier extension paradigms, treating network architecture, lightweight design, scene generalization, and WAM not as isolated research branches, but as a continuous evolutionary process in which VLAs progress from reactive action

generation toward predictive world-action joint modeling. The main contributions of this paper are as follows:

- 1) **VLAs Architectures.** We construct a systematic architectural taxonomy of VLAs, reviewing their core components in terms of vision-language encoding, multimodal backbones, and action decoding, with particular emphasis on summarizing the two major action-generation paradigms of autoregression and diffusion.
- 2) **Advancing VLAs Deployment and Generalization.** From a deployment and generalization perspective, we review the key methods enabling VLAs to move toward real-world application, systematically summarizing frontier progress in lightweight deployment and cross-scene generalization, and analyzing their value and limitations.
- 3) **Boundary and Relationship Clarification of VAM/WAM and VLAs.** we clarify the technical boundary between VAM and WAM and their relationship to VLAs, pointing out that VAM focuses more on the coupling between video temporal dynamics and action sequences, while WAM further emphasizes the joint prediction of future world states and actions, representing an important frontier direction for VLAs moving toward predictive embodied decision-making.
- 4) **Benchmarking VLAs Performance.** We systematically compile and compare the performance metrics of representative VLAs across multiple mainstream simulation datasets, providing empirical evidence for understanding the performance differences, generalization capability, and development bottlenecks of existing methods.
- 5) **Challenges and Future Directions.** Building on the preceding analysis, we reveal the main bottlenecks VLAs face on the path toward real-world deployment and further outline their future development trajectory.

The remainder of this paper is organized as follows: Section II introduces the core architecture of VLAs, their action-generation mechanisms, VAM and WAM, and summarizes related datasets and evaluation benchmarks; Section III discusses model efficiency, scene generalization, and other capability-enhancement methods; Section IV summarizes the main current challenges; Section V summarizes representative applications; Section VI looks ahead to future research directions; and Section VII concludes the paper.

II. VLA Architecture: Core Components and Their Evolution

A. Overview of VLAs

VLAs aim to unify visual perception, language understanding, and robotic action generation within a single decision-making framework, thereby enabling embodied control driven by natural language instructions. A typical VLA pipeline [2], [4], [5], [20], [21] can be summarized

as encoding of vision and language inputs, a multimodal backbone network, and an action-generation module.

1) *Vision Encoder:* The vision encoder is the perceptual front-end of a VLA, responsible for converting observed images into visual feature embeddings that can be used for cross-modal reasoning and action decision-making. Existing VLA systems typically rely on vision foundation models pretrained on large-scale image or image-text paired data, such as ViT [22], CLIP [23], SigLIP [24], and DINOv2 [25], to strengthen the model's ability to represent object semantics, spatial relationships, and scene context. Among these, ViT demonstrates the scalability of modeling sequences of image patches for large-scale visual pretraining; CLIP and SigLIP strengthen the association between visual representations and linguistic semantics through image-text alignment learning; and DINOv2 shows that self-supervised visual pretraining can produce general-purpose features with strong transferability. Overall, these vision foundation models provide important support for open-scene perception and cross-task transfer in VLAs.

2) *Language Instruction:* The language instruction is the semantic interface connecting human task intent with robotic action generation in a VLA, typically used to describe manipulation targets, goal states, spatial relationships, execution order, and behavioral constraints. After being processed by a tokenizer or text encoder, the instruction is converted into language tokens, which together with the visual observation and the robot's proprioceptive state form the conditioning input to the policy model.

3) *LLM Backbone:* In the VLA architecture, the LLM backbone serves as the core semantic modeling module that connects multimodal inputs with action generation. It receives visual representations, language instructions, and historical context, and leverages Transformer-based context modeling to reason about task semantics and interaction states, producing task-conditioned representations for the action decoder. Unlike traditional end-to-end policies, the LLM backbone generally does not directly generate low-level continuous controls.

Large-scale autoregressive pretraining lays the general sequence-modeling foundation for this type of backbone. Pretrained models such as GPT-2 [26], Llama 2 [27], Qwen [28], and Gemma [29] further form transferable linguistic representations and context-modeling capabilities, supporting VLAs in understanding task instructions and organizing multimodal context. In embodied decision-making, Gato [30] uniformly represents text, images, environmental observations, and actions as sequences, and uses a single Transformer to perform multi-task prediction, demonstrating the feasibility of a unified backbone jointly modeling perception and control information. RoboFlamingo [31], in contrast, adopts a pretrained vision-language backbone together with a separate policy head, decoupling multimodal understanding from temporal action generation. Together, these two approaches illustrate two distinct ways of achieving multimodal representation and action generation in embodied models.

4) *Action Decoder*: The action decoding head is a key component of VLAs that bridges multimodal semantic understanding and robotic control. It maps the high-dimensional representations generated by vision-language encoders and multimodal backbones into executable robot actions, such as joint angles, end-effector poses, and gripper states. Since visual and linguistic representations are typically abstract and semantically rich, whereas robotic actions reside in a continuous, low-dimensional, and physically constrained control space, the action decoding head serves as a critical interface for translating high-level semantic representations into low-level physical actions.

In terms of how actions are generated, existing action decoders can broadly be divided into three categories: MLP-based, autoregressive, and diffusion-based. MLP decoders are structurally simple and computationally efficient, typically used to directly regress continuous actions; autoregressive decoders represent actions as tokens or sequence units and generate actions through sequential prediction; diffusion-based decoders model continuous action trajectories through a conditional generative process. Compared with MLP decoders, autoregressive and diffusion-based decoders better reflect the main development trends in large-scale action modeling for current VLAs, so the following analysis focuses primarily on these two categories.

a) *Autoregressive Action Decoders*: Autoregressive action decoders formulate robot action generation as a conditional sequence-prediction problem. The core idea is to discretize or serialize continuous control signals into action tokens, and to predict subsequent actions step by step, in causal order, based on visual observations, language instructions, the robot’s proprioceptive state, and historical actions. By organizing multimodal inputs and action outputs within a unified sequence space, this class of methods can reuse the context-modeling, cross-modal alignment, and semantic-reasoning capabilities of large language models and vision-language models, allowing natural language instruction understanding and robotic action generation to be accomplished within a relatively unified framework.

RT-H [32] builds a hierarchical decision structure through languaged actions and autoregressively generates discrete action tokens in causal order; FAST [33] uses discrete cosine transform and BPE to compress continuous action chunks into shorter token sequences, then predicts subsequent actions step by step based on already-generated tokens, thereby reducing the sequence-modeling overhead of high-frequency control; AR-VLA [34] introduces an independent autoregressive action expert together with a long-term memory mechanism, allowing subsequent actions to continuously depend on historical action context and periodically updated vision-language information, thereby achieving continuous action-stream generation.

Overall, the research focus of autoregressive action decoding is shifting from simple action discretization toward effectively adapting autoregressive sequence modeling to

continuous robotic control, so as to balance semantic modeling, action fidelity, and real-time execution requirements.

b) *Diffusion-Based Action Decoders*: Diffusion-based action decoders formulate robotic action generation as a conditional generative process in continuous space. Such methods are typically conditioned on visual observations, language instructions, and the robot’s proprioceptive state, starting from a noise distribution or an action prior and generating a continuous action sequence through iterative denoising. Compared with autoregressive methods, diffusion-based decoders can directly learn a continuous action distribution and model the multiple plausible trajectories that may exist under the same task condition, making them suitable for tasks such as fine manipulation, dexterous control, and bimanual coordination, which have high requirements for trajectory smoothness and action precision. It should be noted that flow matching also predicts actions through a continuous generative process, but its training objective and generative mechanism are not entirely identical to those of diffusion models; this paper discusses flow matching as a continuous generative action route closely related to diffusion methods.

Existing research primarily improves diffusion-based action generation from the perspectives of decoder structure, model scale, and sampling efficiency. CogACT [35] decouples the vision-language cognitive module from the diffusion action expert to strengthen the functional division between high-level semantic understanding and low-level continuous control; RDT-1B [36] further scales up the diffusion action Transformer and uses a unified action representation to handle heterogeneous robots and bimanual coordination tasks. To address the inference latency caused by multi-step denoising, ManiCM [37] uses consistency distillation to compress action generation into a single or a few inference steps, while CF-VLA [38] adopts a coarse-to-fine flow-generation strategy to improve action initialization and reduce the number of iterations.

Overall, diffusion-based and related continuous generative action decoding is shifting from an emphasis on expressive power for complex action distributions toward jointly balancing large-scale modeling, action-generation quality, and real-time control efficiency.

In summary, VLA action decoding has not yet converged on a unified paradigm; instead, three routes—autoregressive, diffusion-based, and MLP-based—have formed around trade-offs among semantic generalization, control precision, and inference efficiency. Autoregressive decoders make it convenient to reuse the knowledge and reasoning capability of large language models, but are constrained by action discretization error and token-by-token generation latency; diffusion-based decoders and flow matching are better at modeling continuous, multimodal action distributions and can generate smooth, fine-grained trajectories, but incur higher inference overhead; MLP decoders are structurally simple, highly real-time, and suitable for lightweight deployment, but have limited expressive power for complex action distributions. There-

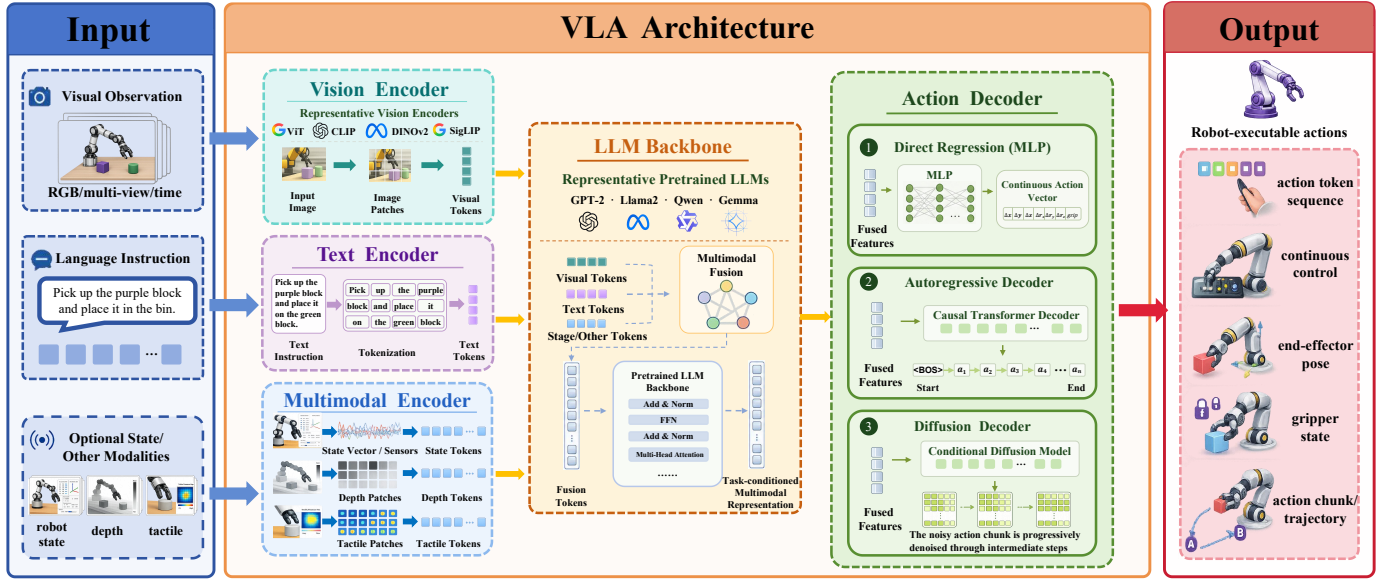


Fig. 2. Architectural overview of VLAs. Encoding: Heterogeneous inputs are encoded and aligned via cross-modal fusion. Multimodal Reasoning: Transformer-based VLM/LLM backbones process fused latent features for high-level decision-making. Action Decoding: Latent representations are mapped into executable robot commands via direct regression, autoregressive prediction, or diffusion/flow matching.

fore, the three approaches are not mutually substitutable, but rather trade-offs suited to different task requirements. Future action decoders may move further toward hybrid and hierarchical designs to jointly balance semantic reasoning, continuous control, and deployment efficiency.

B. VAM: Robot Manipulation Prediction Models that Jointly Model Video Dynamics and Action Sequences

In recent years, with the development of large-scale video generation models and robotic foundation models, VAM has gradually become an important research direction in embodied intelligence. Unlike traditional VLAs, which primarily generate actions directly from the current observation and language instruction, VAM places greater emphasis on learning the temporal structure of task execution from continuous video, linking future visual changes, object-interaction dynamics, and robot actions. Its purpose is not simply to generate future video, but to leverage the spatiotemporal dynamic priors embedded in video models to enhance the temporal continuity and physical consistency of action prediction.

Based on the primary path by which video dynamics participate in action generation, existing VAM approaches can be broadly categorized into two types: coupled video-action generation and cascaded video-planning-to-action conversion. This categorization is not strictly mutually exclusive; for methods that combine both joint modeling and cascaded decoding characteristics, this paper classifies them according to their dominant inference structure.

The first category jointly models future visual dynamics and action sequences within a shared backbone, latent space, or parallel generative branches. UVA [16] learns a shared video-action representation and predicts future observations and actions through independent de-

coding branches, allowing full video generation to be skipped at inference time to reduce computational overhead. VideoVLA [57] transfers a video-generation model to robotic manipulation, jointly predicting future visual changes and action sequences under language and image conditioning. CoVAR [58] achieves bidirectional alignment between video and action features through cross-modal interaction between a video-diffusion branch and an action-diffusion branch. DiT4DiT [61] conditions action generation on intermediate features from the video-denoising process, reflecting a hybrid structure combining joint training with cascaded information transfer.

The second category first generates a future video, a latent visual trajectory, or a goal state, and then converts it into robot control signals via an action decoder or an inverse-dynamics model. VILP [54] predicts a temporally consistent future visual sequence in latent space and uses it as a planning condition for a low-level action policy. ViPRA [55] learns a latent action representation from action-unlabeled videos and then maps it to a continuous control sequence via a decoder. VERA [62] decouples the video planner from a robot-specific inverse-dynamics model, allowing the visual plan to transfer across different robot morphologies. Compared with coupled approaches, this category has a clearer division of labor between modules and makes it easier to separately leverage large-scale video data and limited robotic action data.

Architecturally, VAM typically takes historical images, language instructions, robot state, and historical actions as inputs, models scene evolution via a video-diffusion model or a Video Diffusion Transformer, and then generates control signals via an action-diffusion head, a flow-matching decoder, or an inverse-dynamics module. Its important value lies in its ability to learn dynamic knowledge from in-

TABLE I
REPRESENTATIVE EMBODIED FOUNDATION MODELS WITH DIFFERENT ACTION GENERATION AND PREDICTION PARADIGMS

Model	Year	Key innovation
Autoregressive VLA		
RT-2 [2]	2023	Textualizes actions and fine-tunes on vision-language model to enhance semantic generalization and grounded reasoning.
ECOT [39]	2024	Autoregressively generates a chain of task, perceptual, and motion reasoning before acting.
RT-H [32]	2024	Builds a hierarchy from languaged actions and autoregressively generates discrete action tokens.
CoT-VLA [40]	2025	First autoregressively generates a visual subgoal, then predicts an action chunk in parallel.
ChatVLA [41]	2025	Unifies multimodal understanding and robot control through staged alignment and MoE.
ET-VLA [42]	2025	Adapts to multi-robot collaboration via synthetic continual pretraining and embodied chain-of-thought.
AutoVLA [43]	2025	Uses GRPO to switch between fast and slow reasoning, jointly generating driving reasoning and trajectories.
LoHoVLA [44]	2025	Uniformly generates subtask and action tokens, executing long-horizon tasks via hierarchical closed-loop control.
Libra-VLA [45]	2026	Uses coarse-grained intent to guide diffusion refinement, achieving asynchronous dual-system control.
AR-VLA [34]	2026	Introduces a hybrid KV cache to enable continuous action-stream prediction with long-term memory.
Diffusion-based VLA		
π_0 [5]	2024	Combines a VLM with a flow-matching action expert to generate continuous action chunks.
TinyVLA [46]	2024	Combines a lightweight VLM with a diffusion head, requiring no robot-data pretraining.
DexVLA [47]	2025	Proposes a 1B-parameter diffusion action expert trained with a three-stage embodied curriculum.
SmolVLA [48]	2025	A lightweight VLM with a flow-matching action expert supports asynchronous, low-cost deployment.
DreamVLA [49]	2025	Predicts dynamics, depth, and semantic knowledge to guide diffusion-based action generation.
LLaDA-VLA [50]	2025	Combines a diffusion VLM with action-specific classification and hierarchical decoding.
dVLA [51]	2025	Unifies a discrete diffusion objective to jointly generate multimodal chain-of-thought and actions.
CF-VLA [38]	2026	Replaces pure-noise initialization with an action prior via two-stage "coarse-to-fine" generation.
VLA-Pilot [52]	2026	Optimizes actions at inference time via evolutionary diffusion, enabling fine-tuning-free deployment.
Progressvla [53]	2026	Guides diffusion sampling with progress estimation, optimizing execution and termination of long-horizon tasks.
VAM		
UVA [16]	2025	Jointly learns a video-action latent representation and enables fast action inference via a decoupled diffusion head.
VILP [54]	2025	Generates multi-view-consistent video plans in latent space, mapped to actions via a closed-loop low-level policy.
ViPRA [55]	2025	Jointly predicts future vision and latent motion actions, generating continuous action chunks via a flow-matching decoder.
mimic-video [56]	2025	Conditions a flow-matching action decoder on pretrained video latent representations to improve data efficiency.
VideoVLA [57]	2025	Jointly predicts future video and actions based on a pretrained video generator, enhancing manipulation generalization.
CoVAR [58]	2025	Sets up parallel video and action diffusion branches, cogenerating video-action pairs via Bridge Attention.
S-VAM [59]	2026	Self-distills a multi-step video prior into single-step geometric-semantic foresight for real-time action prediction.
VTAM [60]	2026	Jointly predicts visual-tactile dynamics and enhances contact control with virtual force targets.
DiT4DiT [61]	2026	Jointly trains video and action DiTs, conditioning action generation on video-denoising features.
VERA [62]	2026	Achieves closed-loop cross-embodiment control via a decoupled video planner and a J-IDM (joint inverse dynamics model).
WAM		
UWM [63]	2025	A unified diffusion framework that flexibly achieves policy, dynamics, and video generation via decoupled timesteps.
DreamDojo [64]	2026	Large-scale pretraining on human videos, introducing "continuous latent actions" to address the use of unlabeled data.
Fast-WAM [65]	2026	Remove inference-time imagination, showing that video-modeling synergy at training time is the key to improved performance.
X-WAM [66]	2026	Combines unified 4D modeling with asynchronous denoising, balancing high-fidelity synthesis with real-time execution.
HarmoWAM [67]	2026	Introduces adaptive gating to unify the generalization ability of the world model with the precision of the prediction expert.
OA-WAM [68]	2026	Introduces object-addressable slots, improving the model's interaction robustness in complex scenes.
CKT-WAM [69]	2026	Enables efficient knowledge transfer between heterogeneous models via a compressed interface in text-embedding space.
Latent-WAM [70]	2026	Combines geometric distillation with a causal Transformer for efficient latent-space planning.
τ 0-WM [71]	2026	Builds a unified video-action world-model architecture with a shared backbone for WAM and ACVS.
DreamZero [72]	2026	Jointly generates future video and continuous actions, achieving cross-embodiment transfer with heterogeneous.

ternet videos and human manipulation videos, thereby alleviating the high cost of collecting robotic demonstration data. *mimic-video* [56] leverages the latent representations of pretrained video models to enhance action learning, while *Mitty* [76] explores extracting transferable manipulation knowledge from action-unlabeled human videos.

Overall, VAM extends robotic policies from a static observation-to-action mapping to action generation driven by video dynamics modeling, providing a new technical path for long-horizon manipulation, complex interaction, and cross-embodiment transfer. However, it still faces issues such as the high inference cost of video models, discrepancies between predicted results and executable actions, and the difficulty of stably transferring human video knowledge to robotic platforms. Future research needs to further improve the efficiency of latent-space dynamics modeling and strengthen the physical consistency between

video prediction and closed-loop control.

C. WAM: Predictive Policy Learning Fusing World-State Prediction with Action Generation

With the development of video-generation models, world models, and robotic foundation models, WAM has gradually become an important direction for embodied policy learning. Compared with VLAs, which primarily optimize action generation conditioned on observations, WAM further incorporates future world-state prediction into the action-decision process, enabling models to generate, evaluate, or optimize actions based on anticipated environmental changes. Its core is not merely to generate future images or video, but to establish a connection among world dynamics, action consequences, and control policy, thereby enhancing the robot's ability to model physical change and long-term task processes [17].

WAM and VAM overlap in terms of future visual prediction, but the two differ in emphasis. VAM primarily uses the temporal features learned by video models to enhance action prediction, while WAM is more concerned with the functional coupling between future states and action decisions. Future states can be represented not only as images and video, but also as latent features, object states, 3D geometry, or multimodal physical representations. Whether a method qualifies as WAM does not depend on whether it generates future video, but on whether future-state modeling forms an explicit coupling with action generation.

In terms of the role future states play in the policy, existing WAM approaches first manifest as world-action modeling driven by explicit prediction. DreamZero [72] uses a pretrained video-diffusion model to jointly predict future visual states and action sequences, allowing video dynamic priors to participate in robot control; World-VLA [77] organizes future images and actions within a unified autoregressive sequence, strengthening the association between state evolution and action prediction. This class of methods can provide relatively intuitive future representations, but explicitly generating high-dimensional visual states also increases inference cost and may propagate world-prediction errors into the action output.

To reduce the overhead brought by future generation, recent research has begun to convert world prediction into latent dynamic representations, training-time supervision, or optional inference branches. Fast-WAM [65] retains the future video-modeling objective during training but removes the full video-generation branch at inference time, directly conditioning action prediction on latent world features; GigaWorld-Policy [78] further adopts an action-centered design, using world prediction as auxiliary supervision for training the action policy. HarmoWAM [67] uses an adaptive mechanism to coordinate a world model with strong generalization capability with an action expert oriented toward fine control, while SANTS [79] dynamically adjusts the world-generation and action-execution process based on the current state. Related research shows that the value of WAM is not limited to generating visualizable futures, but is also reflected in strengthening the model's consistency in modeling environmental dynamics and actions through prediction objectives.

At the same time, the research focus of WAM is expanding from holistic visual prediction toward structured world representations and world-action interaction. OA-WAM [68] adopts an object-addressable representation, separately modeling the robot, target objects, and their time-varying states, enhancing target localization and interaction robustness in complex scenes; X-WAM [66] fuses multi-view RGB-D information, extending world prediction to a 4D representation that includes spatial geometric change; GeoSem-WAM [80] further jointly leverages geometric structure and semantic cues to improve action-conditioned world modeling capability. DAWN [81] enables world prediction and action generation to recursively refine each other during inference, extending WAM to

interactive autonomous-driving planning; WorldVLN [82] uses an autoregressive video backbone to predict short-horizon latent world-state transitions and decodes them into executable waypoints, continuously updating its prediction context under real observational feedback, demonstrating a closed-loop implementation of WAM in aerial vision-language navigation.

Overall, WAM extends VLAs from observation-conditioned action generation to predictive decision-making constrained by world evolution. Its development focus is shifting from explicitly generating a complete future state toward efficient latent modeling, object- and geometry-structured representation, and bidirectional world-action interaction. However, predicted states may deviate from the true physical process and further mislead action decisions; high-dimensional world generation also struggles to meet the demands of high-frequency closed-loop control. In addition, existing evaluation methods still struggle to simultaneously measure the fidelity of future states, the physical plausibility of action consequences, and the executability of control signals. Future research needs to reduce the propagation of world-prediction error into action policies and establish a unified evaluation system that jointly considers prediction quality, control efficiency, and task success rate.

D. Datasets

Existing datasets and benchmarks for VLAs generally show a trend evolving from single-task manipulation toward multi-task generalization, language-conditioned control, long-horizon planning, robustness evaluation, and multi-robot-embodiment transfer. Early datasets mostly focused on multi-task reinforcement learning and skill transfer for robots in simulated environments, such as Meta-World; subsequently, benchmarks such as LIBERO and CALVIN further introduced natural language instructions, demonstration data, and long-horizon tasks, requiring models not only to complete low-level action control but also to understand task semantics and compose multiple subtasks. More recently, LIBERO-Plus and RoboTwin 2.0 have further focused on the robustness and generalization of VLAs in real, complex environments, examining the impact of factors such as camera viewpoint, object layout, robot initial state, lighting, background, and bimanual coordination on model performance.

1) *Meta-World*: Meta-World is a simulated robotic manipulation benchmark containing 50 distinct manipulation tasks with shared object and interaction structures. It is designed to evaluate the capabilities of reinforcement learning and imitation learning algorithms in multi-task learning, task transfer, and generalization to unseen tasks. Meta-World defines several evaluation settings, including the ML series for meta-learning scenarios and the MT series for multi-task learning scenarios. The ML settings evaluate the ability to adapt to new tasks with limited demonstrations, while the MT settings focus on learning multiple manipulation tasks within a shared environment.

Due to its standardized simulation environments and diverse task collection, Meta-World has been widely adopted for evaluating robotic manipulation policies [85].

2) *LIBERO*: LIBERO is a lifelong robot learning benchmark designed to study knowledge transfer in robotic manipulation. It focuses on transferring different types of knowledge, including task semantics, object configurations, and manipulation procedures, across sequential learning settings. LIBERO provides a procedural task-generation pipeline and contains 130 manipulation tasks organized into four suites: LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-100. LIBERO-Spatial evaluates spatial knowledge transfer, LIBERO-Object focuses on object-level knowledge transfer, LIBERO-Goal studies goal-related knowledge transfer, and LIBERO-100 contains a larger collection of diverse manipulation tasks for lifelong learning evaluation. In addition, LIBERO provides high-quality human teleoperation demonstrations, making it suitable for imitation learning, lifelong robot learning, and vision-language-action model evaluation [8].

3) *LIBERO-Plus*: LIBERO-Plus is a robustness evaluation framework built upon the LIBERO benchmark to analyze the generalization capability of vision-language-action models under environmental perturbations. It introduces diverse perturbation factors, including object layout, camera viewpoint, robot initial state, language instructions, lighting conditions, background texture, and sensor noise. By evaluating models under these controlled distribution shifts, LIBERO-Plus aims to measure the robustness of robotic policies beyond standard benchmark settings [86].

4) *RoboTwin 2.0*: RoboTwin 2.0 is a data-generation and evaluation platform designed for bimanual robotic manipulation. It provides a collection of manipulation tasks covering various interaction scenarios, including object pick-and-place, bimanual handover, stacking, sorting, articulated-object manipulation, tool use, and fine-grained interaction. The benchmark supports multiple robot platforms and improves data diversity through variations in scene configurations, lighting conditions, backgrounds, tabletop heights, and language instructions. RoboTwin 2.0 introduces Easy and Hard evaluation settings to measure policy performance under standard environments and more challenging conditions with increased environmental variations, respectively [10].

5) *CALVIN*: CALVIN is a simulated benchmark for language-conditioned long-horizon robotic manipulation. It contains 34 composable manipulation tasks involving interactions such as block manipulation, drawer opening, sliding-door operation, and light-switch control. The benchmark evaluates long-horizon task execution by composing basic tasks into sequential instruction chains, requiring robots to complete multiple subtasks according to natural-language commands. CALVIN provides multimodal observations and supports different action representations, including absolute Cartesian poses, relative Cartesian displacements, and joint actions. Therefore, it

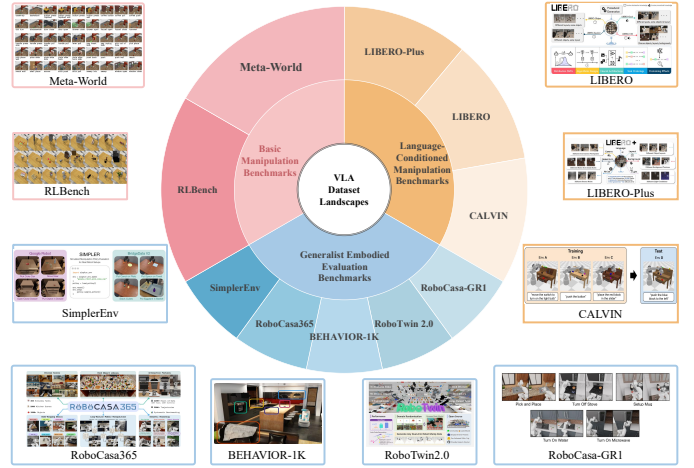


Fig. 3. Landscape of dataset and evaluation benchmarks for generalist VLAs. Overview of VLA evaluation frameworks categorized into basic manipulation benchmarks, language-conditioned manipulation benchmarks, and generalist embodied evaluation benchmarks. Outer visual callouts illustrate representative simulation setups and environment dynamics.

has been widely used to evaluate language-conditioned policy learning, compositional generalization, and long-horizon manipulation capabilities [87].

In addition to the above benchmarks, RoboCasa [9], RoboCasa-GR1, RLBench [88], BEHAVIOR-1K [89], and SimplerEnv [90] provide complementary resources for VLA training and evaluation. RoboCasa provides large-scale household manipulation environments with diverse robotic policies. RoboCasa-GR1 focuses on evaluating robot policies under standardized environments and evaluation protocols. RLBench provides a collection of simulated manipulation tasks with multimodal observations and demonstration data, and has been widely used in reinforcement learning, imitation learning, and multi-task policy evaluation. BEHAVIOR-1K targets long-horizon household activities and emphasizes complex object interactions and task-level reasoning. SimplerEnv provides a unified simulation-based evaluation environment for assessing the transferability of real-robot policies under simulated settings. Together, these datasets and benchmarks extend VLA evaluation across household manipulation, long-horizon activities, standardized manipulation tasks, and sim-to-real transfer.

III. Deployment and Generalization

A. Model Efficiency

However, as model parameter scale grows and multi-modal fusion becomes more complex, VLAs increasingly face problems such as high computational overhead and low inference efficiency in real-world deployment. RT-2 points out that its high computational cost limits real-time inference in high-frequency control scenarios, and that optimization via methods such as quantization and distillation is urgently needed [2]; OpenVLA points out

that most existing VLAs are closed-source and lack deployment and adaptation practices for new robots, environments, and tasks, particularly on consumer-grade hardware [4]. Subsequent research has repeatedly emphasized issues such as high training and inference overhead and insufficient real-time performance [46], [91], [92]. Based on this, this section reviews three main directions—visual token pruning, quantization, and distillation—which aim to reduce computational overhead while preserving model performance and generalization ability as much as possible.

1) *Visual Token Pruning*: Visual token pruning reduces self-attention overhead by removing redundant visual tokens from Transformers. Unlike pruning methods for general VLMs that mainly rely on visual saliency, token selection in VLAs must also consider action relevance, temporal dependency, and geometric structure to avoid discarding information critical to manipulation. Accordingly, recent studies have gradually shifted from perception-driven pruning toward action- and task-aware strategies.

Training-free methods can be directly integrated into existing VLAs without additional optimization. VLA-IAP [93] preserves key manipulation structures using geometric priors and dynamically adjusts pruning strength. DTP [94] removes visual tokens from task-irrelevant regions to refine visual attention. ADP [95] adaptively adjusts the token retention ratio according to action trajectories. SpecPrune-VLA [96] performs dynamic pruning based on action temporal information and layer-wise token importance. These methods mainly exploit action, motion, or geometric priors to improve token selection while reducing computation without retraining.

Training-based methods learn task-relevant token selection through fine-tuning or end-to-end optimization. DeepVision-VLA [97] removes irrelevant tokens using an action-guided visual pruning module. BFA++ [98] reduces spatial and multi-view redundancy through intra-view and inter-view predictors. ETA-VLA [99] compresses cross-frame tokens by combining text relevance with temporal consistency. LightVLA [100] adopts differentiable pruning to learn token retention strategies. CogVLA [101] removes instruction-irrelevant information through progressive sparsification, while VLA-Pruner [102] evaluates and compresses redundant tokens according to action relevance. Compared with training-free methods, these approaches can learn finer-grained task-specific token importance through optimization.

System-level co-optimization further combines token pruning with other acceleration techniques. SP-VLA [103] integrates token pruning with dynamic model scheduling. AC²-VLA [104] jointly performs feature reuse, token pruning, and selective component execution. SQAP-VLA [105] combines quantization and pruning for joint acceleration. These methods extend beyond token compression toward multidimensional computational optimization, further improving end-to-end inference efficiency.

As a whole, existing visual token pruning methods

mainly follow three directions: training-free pruning, training-based pruning, and system-level co-optimization. They aim to reduce the computational cost of VLAs while largely preserving task performance, thereby supporting efficient deployment on resource-constrained platforms.

2) *Quantization*: Quantization reduces memory usage and inference latency by mapping model weights and activations to low-bit representations, making it an important technique for deploying VLAs on resource-constrained platforms. However, due to the characteristics of continuous closed-loop control, heterogeneous multimodal representations, and the sensitivity of action outputs to numerical precision, directly applying LLM-oriented quantization methods to VLAs may lead to substantial performance degradation.

For post-training quantization, QuantVLA [106] achieves training-free quantization through multi-scale calibration. DA-PTQ [107] combines cross-space representation compensation with mixed-precision allocation to mitigate kinematic drift caused by low-bit quantization. QVLA [108] performs channel-wise quantization based on action-space sensitivity and jointly optimizes quantization and pruning. LiteVLA-Edge [109] further integrates low-bit quantization with a lightweight inference framework, enabling real-time VLA deployment on Jetson Orin with ROS2. These methods mainly rely on control-aware calibration and precision allocation to reduce quantization errors without retraining.

For extreme low-bit quantization, HBVLA [110] adopts policy-aware post-training quantization to preserve control performance under aggressive compression, while BitVLA [92] employs a native ternary backbone and quantization distillation to reduce model storage. These methods explore post-training quantization and native low-bit training as two complementary directions for extreme compression.

For dynamic quantization and quantization-aware training, DyQ-VLA [111] dynamically adjusts quantization precision according to state sensitivity. SQIL [112] improves low-bit control accuracy by assigning greater training weights to critical states, while QAIL [113] enhances robustness to quantization errors through quantization-aware fine-tuning. These methods mitigate the impact of quantization errors on action decisions through dynamic precision allocation and training-time robustness optimization.

Overall, VLA quantization is evolving from generic model compression toward control-aware optimization, with increasing emphasis on dynamic precision, joint optimization with other acceleration techniques, and deployment on real robotic systems.

3) *Distillation*: Distillation transfers knowledge from a large teacher model to a lightweight student model, enabling model compression while preserving multimodal understanding and action-generation capabilities. Compared with pruning and quantization, distillation can transfer semantic, structural, and action-related knowledge simul-

taneously, making it particularly suitable for capability transfer in embodied tasks.

For semantic and action distillation, ActDistill [114] models the action prediction process through graph structures and introduces dynamic routing for action-aware computation-path selection. VITA-VLA [115] transfers the execution capability of a small action expert to a vision-language model through action-space mapping and two-stage fine-tuning. These methods rely mainly on action-related supervision to preserve task understanding and execution capabilities after compression.

For structural distillation, Shallow- [116] reduces the Transformer depth of both the vision-language backbone and the Flow Matching action head. DySL-VLA [117] learns to dynamically skip redundant layers through distillation. AC²-VLA [104] combines cognitive reuse, token pruning, and selective layer execution with action-guided self-distillation. DualVLA [118] employs dual-teacher distillation to balance reasoning generalization and action accuracy, while HY-Embodied-0.5 [119] uses online policy distillation to transfer the capabilities of a large model to an edge-oriented variant. These methods mainly compress network depth and inter-layer information flow to reduce inference costs.

For Flow Matching self-distillation, SnapFlow [120] constructs distillation signals from the model’s own velocity predictions and compresses multi-step iterative denoising into a single forward pass, reducing dependence on external teacher models and improving action-generation efficiency.

In general, VLA distillation is evolving from simple action imitation toward semantic transfer, structural compression, and self-distillation, providing more flexible compression strategies for efficient deployment.

B. Scene Generalization

To address the performance degradation of VLAs in out-of-distribution environments, existing research has mainly formed three technical routes: spatial-geometric perception reduces a model’s reliance on surface features such as background by strengthening its understanding of object position and spatial relationships; world models improve a policy’s anticipatory capability by predicting environmental change and future states; and inference-time intervention combined with multi-skill composition enhances adaptation to new scenes and complex tasks without requiring model retraining.

1) *Spatial-Geometric Perception and Object-Centric Representation*: Traditional vision encoders are mainly optimized for semantic invariance and therefore have limited sensitivity to fine-grained spatial displacement and geometric structure. Recent studies address this limitation through three main directions: geometric-prior injection, multi-layer and multi-view alignment, and object-centric grounding.

For geometric-prior injection, LaMP [121] uses 3D scene flow as an action-generation condition to provide motion

priors for action decoding. Robot-DIFT [122] distills geometric knowledge from diffusion models into the vision encoder, while GLaD [123] further extends geometric alignment to LLM hidden states to improve spatial reasoning. These methods enhance spatial modeling by introducing geometric information at either the action or encoder level.

For multi-layer and multi-view alignment, ROCKET [124] fuses spatial information from 3D vision foundation models through multi-layer feature alignment. PEAfowl [125] builds geometrically consistent cross-view representations and uses depth distillation to improve spatial perception in bimanual manipulation. PokeVLA [126] introduces geometry-aware tasks during pretraining, achieving strong spatial generalization with a relatively small model. These methods exploit multi-layer and multi-view information to improve spatial representation.

For object-centric grounding, VLAs often struggle to stably bind language instructions to scene entities under distractors or ambiguous instructions. ProGAL-VLA [127] enhances language–entity alignment through a 3D object-centric graph, while TAG [128] highlights target-relevant information and suppresses distractors through goal-erased comparison. The two methods improve grounding through structured training and lightweight inference-time intervention, respectively.

These approaches improve VLA generalization in complex manipulation scenarios by strengthening geometric modeling, cross-view representation, and language–entity grounding.

2) *World-Model-Driven Anticipatory Prediction*: Traditional VLAs are essentially reactive controllers that establish a static mapping directly from observation to action, with a decision process that lacks active modeling of the environment’s physical dynamic evolution. Introducing a World Model (WM) is a core idea for overcoming this limitation: by learning spatiotemporal physical priors from massive video data, a policy can gain, at decision time, an expectation of “how an action will change the state of the world.” Recent research has further evolved into world-action models, deeply integrating physical prediction with action generation. Around how the world model is introduced and how it is used at inference time, two current routes exist.

World-model-based methods. This route does not explicitly generate future visual observations, but instead injects video-prediction capability into a VLA’s encoding process in representational form. JEPA-VLA [129] and VLA-JEPA [130] are both based on the JEPA framework, differing mainly in how they fuse information: the former injects pretrained video-prediction embeddings into an existing VLA architecture via adaptive fusion; the latter adopts a strict leakage-free latent-state prediction mechanism, where the student path receives only the current observation rather than future frames, thereby forcing the model to learn genuine dynamic abstraction in latent space. AtomVLA [131] applies a predictive WM at the post-training stage, using a large language model

TABLE II
Representative Works on LightWeight VLAs

Method	Year	Backbone	Core idea
Token pruning			
BFA++ [98]	2026	Adapter	Dynamic hierarchical token pruning based on two-level intra- and inter-view importance predictors.
ETA-VLA [99]	2026	Vicuna-7B	Uses text and temporal cues to sparsely aggregate historical multi-view tokens in the LLM.
VLA-Pruner [102]	2026	Adapter	Prunes tokens using semantic and action relevance, retaining key tokens through merge-then-filter.
DeepVision-VLA [97]	2026	Qwen3-VL (4B)	Uses VL-MoT and shallow attention pruning to improve visual grounding.
LightVLA [100]	2025	Llama 2 7B	Uses differentiable pruning for adaptive token retention without manual thresholds.
ADP [95]	2025	Llama 2 7B	Adaptively adjusts token retention using text selection and action-aware gating.
SpecPrune-VLA [96]	2025	Adapter	Uses action-history consistency and hierarchical importance for action- and layer-level pruning.
CogVLA [101]	2026	Llama 2 7B	Applies three-stage sparsification with instruction-guided routing and coupled attention.
SP-VLA [103]	2025	Llama 2 7B	Combines action-aware scheduling with spatiotemporal-semantic token pruning.
AC ² -VLA [104]	2026	Prismatic-7B	Adaptively performs cognitive reuse, token pruning, and component selection using action context.
SQAP-VLA [105]	2025	Adapter	Jointly optimizes quantization and token pruning for training-free acceleration.
DTP [94]	2026	Adapter	Dynamically removes task-irrelevant tokens to correct attention drift.
VLA-IAP [93]	2026	Adapter	Combines semantic-motion alignment with geometric priors for training-free pruning.
Quantization			
QuantVLA [106]	2026	Adapter	Compresses language and action modules via selective PTQ and scale calibration.
DA-PTQ [107]	2026	LLaMa 2	Stabilizes PTQ trajectory drift with compensation and mixed precision.
BitVLA [92]	2026	BitNet	Builds a native 1-bit VLA and compresses the vision encoder via quantize-then-distill.
SQAP-VLA [105]	2025	Adapter	Alleviates conflict between quantization and token pruning via quantization-aware pruning.
DyQ-VLA [111]	2026	Llama 2 7B	Dynamically allocates bit-width using kinematic sensitivity.
LiteVLA-Edge [109]	2026	SmolVLM-256M	Uses FP32 fine-tuning and 4-bit quantization for Jetson deployment.
HBVLA [110]	2026	Llama 2 7B	Achieves 1-bit PTQ by combining a policy-aware Hessian with Haar-domain binarization.
QVLA [108]	2026	Llama 2 7B	Performs channel-wise mixed-precision quantization based on action-space sensitivity.
SQIL [112]	2025	Llama 2 7B	Strengthens key states via saliency-aware QAT to recover low-bit policy performance.
QAIL [113]	2024	Llama 2 7B	Improves low-bit policy robustness through quantization-aware imitation learning.
Distillation			
HY-Embodied-0.5 [119]	2026	Hunyuan-1.8B	Distills large-model embodied reasoning capability into an edge-friendly small model.
SnapFlow [120]	2026	Adapter	Self-distillation compresses multi-step denoising into single-step generation.
DySL-VLA [117]	2026	Adapter	Dynamically skips redundant layers to accelerate key-action inference.
Shallow- π [116]	2026	Gemma-2.6B	Distills the VLM backbone and action head from deep to shallow.
AC ² -VLA [104]	2026	Prismatic-7B	Adaptively performs cognitive reuse, token pruning, and component selection using action context.
DualVLA [118]	2025	Prismatic-7B	Balances reasoning and execution via a reasoning teacher and an action teacher.
ActDistill [114]	2026	Llama 2	Distills action-relevant computation paths combined with dynamic routing.
VITA-VLA [115]	2025	Qwen-2.5-7B	Distills a small action expert into the VLM to improve action-generation efficiency.

to decompose high-level demonstrations into fine-grained atomic subtasks, and using a pretrained WM to score candidate action chunks in latent space. Its core contribution lies in mitigating long-horizon error accumulation through task decomposition and latent-space verification, forming an orthogonal complement to the aforementioned work targeting visual-scene perturbation.

World-action-model-based methods. WAM incorporates video tokens and action tokens into a unified generative framework, allowing the policy to decode actions synchronously while predicting future world-state evolution. OA-WAM [68] adopts an object-centric representation that decouples object identity from dynamic state, mitigating semantic confusion in global image prediction; HarmoWAM [67] combines the two strategies of "imagine before acting" and joint modeling, coordinating generalization ability and control precision through adaptive gating.

3) Inference-Time Adaptive Intervention and Multi-Skill Compositional Generalization: While the preceding methods mainly improve generalization during training, this line of research focuses on inference-time adaptation with frozen pretrained policies. It enhances OOD adaptation without parameter updates through visual intervention, online memory, and skill orchestration.

For inference-time visual intervention, CGVD [132] suppresses semantic noise through target-distractor separation and image inpainting, improving visual attention at

the input level. AFI [133] combines 3D spatial affordances with proprioceptive feedback to detect execution stagnation and redirect the robot toward high-affordance regions. These methods improve inference-time adaptation through perceptual refinement and execution-level intervention, respectively.

For online memory and dynamic orchestration, SOMA [134] combines dual-memory RAG with an LLM orchestrator to retrieve behavioral priors and perform modular interventions without parameter fine-tuning. SPR [135] dynamically converts language instructions into spatial subgoals and returns to recoverable states when execution stalls. SOMA provides greater system-level flexibility, whereas SPR adopts a lighter and more direct intervention strategy.

For dynamic routing and multi-skill integration, ActThink-Abstain [136] dynamically switches among direct execution, further reasoning, and abstention according to physical and semantic anomalies, improving decision safety in complex scenarios. MergeVLA [137] enables multi-skill integration through mergeable task structures, while StarVLA- [138] shows that a relatively simple unified architecture can also achieve strong multi-task generalization.

Collectively, these methods enhance the online adaptability, robustness, and multi-task generalization of frozen VLAs through inference-time intervention, external mem-

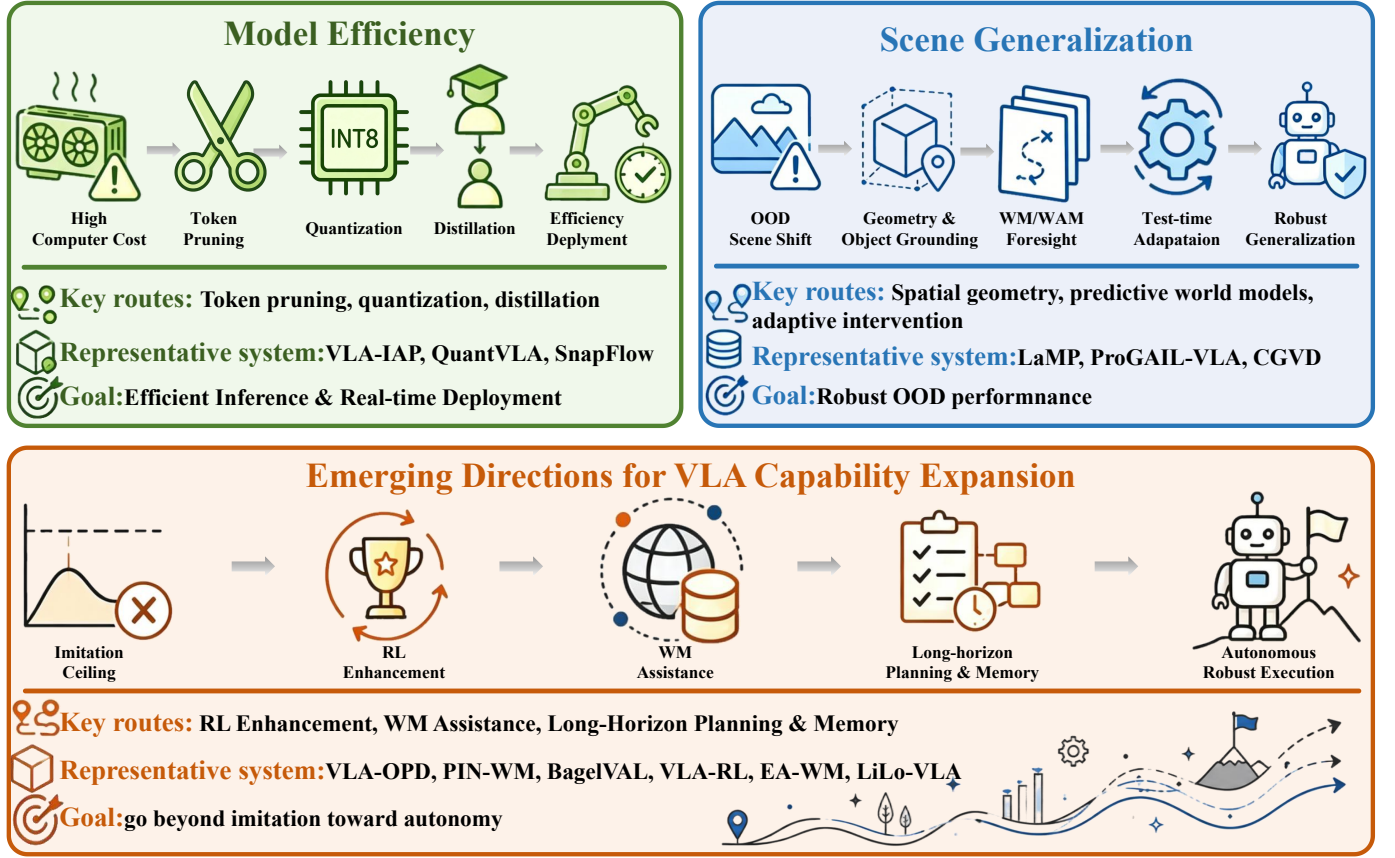


Fig. 4. Taxonomic overview of technical routes for VLAs efficiency, scene generalization, and capability expansion. It outlines the core research directions of VLAs: enhancing model efficiency via pruning, quantization, and knowledge distillation; overcoming scene generalization bottlenecks by leveraging spatial geometric perception and adaptation; and exploring emerging capability expansion through reinforcement learning, world models, and long-horizon tasks. The key technical pathways, representative systems, and target goals for each direction are comprehensively summarized.

ory, dynamic routing, and modular skill composition.

C. Capability Extension and Emerging Research Directions for VLAs

Although lightweight design and scene generalization lay the foundation for efficient VLA deployment, relying solely on the imitation-learning paradigm still has fundamental limitations for enabling robots to achieve autonomous, robust, long-term operation in the real world: a policy cannot exceed the distributional boundary of its demonstration data, lacks explicit reasoning about future states, and is prone to error accumulation and instruction drift as the task's time horizon increases. To address this, recent research has built capability-enhancement frameworks along three complementary dimensions: reinforcement learning (RL) breaks through the performance ceiling of imitation learning via interaction-driven optimization; world models (WM) support future-state prediction and policy improvement via internal modeling of environmental dynamics; and long-horizon reasoning improves cross-stage execution consistency by introducing explicit planning structures and memory mechanisms. Notably, these three are deeply interconnected: world models provide low-cost simulated environments for RL, RL in turn drives policy exploration within world models,

and together the two provide reasoning support for long-horizon tasks.

1) *Reinforcement-Learning-Based VLAs*: Although pre-training and supervised fine-tuning have significantly improved VLA performance, imitation learning remains constrained by the quality and coverage of demonstration data, making it prone to error accumulation and insufficient adaptation in out-of-distribution scenarios. Reinforcement learning has therefore emerged as an important direction for VLA post-training, enabling further policy improvement through interaction-driven optimization. Existing studies mainly focus on optimization objectives, exploration efficiency, data generation, task specialization, and system scalability.

For optimization objectives, VLA-OPD [140] adopts online policy distillation, where a teacher model provides fine-grained supervision over trajectories generated by the student. ARFM [141] formulates an offline reinforcement learning objective for Flow Matching VLAs. VLA-RL [142] learns process-level rewards from pseudo-rewards, while SRPO [143] uses successful trajectories to provide progress supervision for failed trajectories. These methods mainly extend the optimization signal from terminal rewards to process-level feedback.

For exploration efficiency, IG-RFT [144] dynamically

TABLE III
Representative Works on Scene Generalization

Method	Year	Benchmark	Success rate	Core idea
OpenVLA [4]	2024	LIBERO-Plus	15.6	Formulates robotic control as VLM autoregressive prediction via action tokens.
WorldVLA [77]	2025	LIBERO-Plus	25.0	Unifies VLA and world modeling to learn physical environment dynamics.
JEPa-VLA [129]	2026	LIBERO-Plus	25.6	Injects video-prediction embeddings, enhancing temporal dynamics modeling.
UniVLA [139]	2025	LIBERO-Plus	42.9	Learns task-centric latent action representations for cross-embodiment.
π_0 [5]	2025	LIBERO-Plus	53.6	Adopts a dedicated action expert with flow matching to generate action chunks.
π_0 -Fast [33]	2025	LIBERO-Plus	61.6	Employs FAST action tokenization for single-pass action generation.
SPR [135]	2026	LIBERO-Plus	71.8	Enables robust execution via progress monitoring and rollback.
MergeVLA [137]	2026	LIBERO-Plus	72.4	Integrates multiple skills via a mergeable architecture.
LaMP [121]	2026	LIBERO-Plus	79.3	Injects 3D scene flow priors into action decoding.
VLA-JEPa [130]	2026	LIBERO-Plus	79.5	Learns dynamic representations via latent-state prediction for pretraining.
StarVLA- α [138]	2026	LIBERO-Plus	79.7	Uses Qwen3-VL and lightweight MLP for strong generalization.
ROCKET [124]	2026	LIBERO-Plus	81.7	Injects 3D geometry via residual alignment, improving spatial generalization.
PokeVLA [126]	2026	LIBERO-Plus	83.5	Injects geometric representations via two-stage training, enhancing manipulation.
OA-WAM [68]	2026	LIBERO-Plus	83.9	Uses object slots to strengthen target binding and robustness.
ProGAL [127]	2026	LIBERO-Plus	85.5	Strengthens language-scene binding via entity graphs and contrastive alignment.
TAG [128]	2026	LIBERO-Plus	87.2	Reinforces target evidence via a goal-erasure residual, suppressing distractor bias.

adjusts exploration intensity according to interaction states and combines trajectory-level and subtask-level rewards. RL [145] formulates the Flow Matching denoising process as a Markov decision process, while FASTER [146] pre-filters candidate actions during action denoising. These methods improve policy exploration and inference efficiency through structured modeling of the interaction or denoising process.

For RL-driven data generation, DLR [147] generates more diverse successful trajectories by promoting behavioral diversity. PLD [148] uses residual reinforcement learning to identify failure regions of a base policy and distills the resulting recovery experience back into a general-purpose VLA. This direction further extends reinforcement learning from policy optimization to data generation.

For generalization and task specialization, SimpleVLA-RL [149] improves generalization through trajectory sampling and enhanced exploration. GR-RL [150] specializes a general-purpose VLA into a high-precision manipulation policy. RPD [151] combines online reinforcement learning with behavior cloning to accelerate policy learning, while LongNav-R1 [152] improves credit assignment in long-horizon navigation through multi-turn interaction and horizon-adaptive optimization.

For system scalability, RL-VLA³ [153] adopts a fully asynchronous distributed training framework and improves coordination among simulation, inference, and training through dynamic batching and environment sharding.

Reinforcement learning is evolving from a standalone post-training optimization technique into a broader framework that encompasses policy optimization, data generation, task specialization, and system scaling.

2) *World-Model-Based VLAs*: Many end-to-end VLAs rely primarily on current observations or short observation histories for action prediction, lacking explicit modeling of environmental dynamics. As a result, they are prone to prediction errors and error accumulation in long-horizon tasks. World models address this limitation by construct-

ing predictive androllable internal representations of the environment, allowing VLAs to estimate future state changes before execution and use these predictions to support action selection and policy optimization. Existing research mainly focuses on environmental dynamics representation, policy correction, and supervision generation.

For environmental dynamics representation, current methods mainly include latent-space modeling, explicit physical modeling, and structured representations. LaDi-WM [154] predicts environmental changes using semantically aligned latent features to improve generalization in dynamics prediction, while DDP-WM [155] separately models interacting agents and background dynamics to reduce redundant computation. ContactGaussian-WM [156] jointly models visual appearance, collision geometry, and contact dynamics, whereas PIN-WM [157] uses differentiable physical simulation to estimate key dynamics parameters, enhancing physical prediction and Sim2Real transfer. H-WM [158], EA-WM [159], and OSVI-WM [160] further construct structured representations from the perspectives of task states, kinematic relations, and latent trajectories, respectively, providing more explicit intermediate predictions for long-horizon execution and generalization to unseen tasks.

For policy correction and supervision generation, the imagination and rollout capabilities of world models are further exploited to generate training data and recovery experience. Hi-WM [161] allows policies to repeatedly execute in imagined environments and generates recovery trajectories through state rollback and human correction. WM-Dagger [162] further automates the synthesis and filtering of corrective data to reduce human intervention. WoVR [163] uses the world model as an interactive environment for reinforcement learning and mitigates error accumulation during long-horizon imagination through action-conditioned prediction, keyframe initialization, and joint policy-world-model updates. These methods extend world models from pure environment predictors into platforms for data generation and policy optimization.

World models are evolving from future-state prediction modules into integrated mechanisms that support environment modeling, policy correction, supervision generation, and interactive optimization, providing stronger dynamic reasoning capabilities for long-horizon VLA decision-making.

3) *VLAs for Long-Horizon Tasks*: Long-horizon failures are usually not caused by a single bottleneck, but are jointly amplified by multiple types of deficiency within the execution loop: task goals are difficult to ground consistently across stages; key historical states are lost within a Markovian observation window; local errors accumulate over time, leading to irreversible trajectory deviation; and there is a lack of reliable detection and recovery mechanisms after execution failure. Recent research addresses these root causes of failure through systematic exploration along four dimensions: hierarchical planning, persistent memory, adaptive recovery, and predictive unified reasoning.

For hierarchical planning, VLAs-as-Tools [164] coordinates a high-level VLM with a specialized VLA for task planning and execution. Anticipation-VLA [165] recursively generates subgoals based on the current state. LoHo-Manip [166] uses a receding-horizon manager to predict task progress and future trajectories, while LiLo-VLA [167] separates global motion planning from local interaction to improve compositional generalization. These methods reduce long-horizon execution complexity through hierarchical coordination between planning and control.

For persistent memory, HELM [168] builds a memory-recovery loop through keyframe retrieval, state verification, and rollback. VQ-Memory [169] compresses historical proprioceptive states into discrete memory representations. TempoFit [170] reuses historical K/V caches to extend temporal context, while KC-VLA [171] dynamically retrieve and reinject key historical frames into the model. These methods mitigate long-term dependency issues by preserving and selectively recalling task-relevant historical information.

For adaptive recovery, Critic in the Loop [172] uses a visual Critic to detect task completion, failure, or stagnation and trigger VLM replanning. RoboClaw [173] combines forward manipulation with reverse recovery to form a self-resetting loop. Goal2Skill [174] integrates VLM planning, outcome verification, and VLA execution into a continuous recovery framework. These methods improve anomaly detection and online correction after failures, although they also increase system complexity.

For predictive unified reasoning, BagelVLA [175] jointly models language planning, visual prediction, and action generation, incorporating future-state prediction into execution. AtomVLA [131] supports long-horizon decision-making through atomic subtask decomposition and world-model-based action scoring. Compared with modular pipelines, these methods can reduce information fragmentation across components, but they still face challenges in joint training and real-time inference efficiency.

Long-horizon VLA research mainly improves execution consistency and robustness through hierarchical task decomposition, persistent memory, closed-loop recovery, and future-state prediction.

IV. Core Challenges Facing VLAs

VLAs have already made significant progress on short-horizon, standard tasks, but there remains a considerable gap before they become generalizable, deployable, continuously evolving general-purpose embodied intelligence systems. These gaps correspond to structural bottlenecks along seven dimensions: model capability, deployment efficiency, data infrastructure, physical understanding, task reasoning, system architecture, and data sourcing. These seven categories of challenges cover the main limitations current VLAs face in terms of efficiency, generalization, reasoning, interaction, and deployment, together reflecting the key problems that still need to be solved on the path from experimental settings to real-world application.

A. Limitations in Cross-Scene Generalization

VLAs typically perform stably within the training distribution, but performance often drops markedly when facing unseen tasks, scenes, robot embodiments, and long-tail perturbations. An important reason for this is that models tend to learn appearance statistics and trajectory regularities present in the training data, without forming transferable knowledge of object relations, spatial structure, and action. Existing vision encoders typically encode target entities, background texture, and camera viewpoint together into implicit features, making action prediction susceptible to appearance change; when object position, scene layout, or robot embodiment changes, the correspondence between language instruction, visual target, and action can also become unstable. Under instruction rephrasing, target distraction, or ambiguous expression, some models still tend to rely on visual cues and trajectory priors from training rather than fully exploiting linguistic information. Therefore, the generalization bottleneck of VLAs concerns not only visual encoding capability, but also insufficient object-relation modeling, spatial-language grounding, and cross-embodiment action alignment.

B. Fragility in Long-Horizon Task Execution

The difficulty of long-horizon manipulation tasks lies not only in insufficient context length, but also in the fact that most models lack a verifiable representation of task progress and a closed-loop recovery mechanism. Most VLAs are primarily reactive mappings from observation to action, unable to continuously self-evaluate their execution state; failures at key nodes can propagate irreversibly along the execution chain. At the same time, task progress is treated as a continuous action stream rather than a discrete state machine, making it difficult for a model to judge whether a subtask has been completed or to invoke key historical states in cross-stage decision-making. Although methods such as subtask decomposition and

TABLE IV
Representative Works on Other Directions of VLA

Model	Year	Key innovation
Reinforcement Learning		
FASTER [146]	2026	Pre-filters high-value actions during denoising to reduce sampling cost.
VLA-OPD [140]	2026	Uses teacher supervision to combine SFT efficiency with RL robustness.
IG-RFT [144]	2026	Uses interaction-guided exploration and dense rewards for long-horizon manipulation.
RL-VLA ³ [153]	2026	Improves the system throughput of VLA-RL via asynchronous distributed training.
GR-RL [150]	2025	Combines demonstration filtering, action augmentation, and online RL for fine-grained long-horizon tasks.
DLR [147]	2025	Uses multi-mode RL to generate diverse successful trajectories, expanding VLA pretraining data.
PLD [148]	2025	Discovers failure regions via residual RL and distills recovery trajectories back into the VLA.
π RL [145]	2026	Uses Flow-Noise and Flow SDE to address likelihood computation and exploration.
SimpleVLA-RL [149]	2025	Builds a scalable online RL framework, improving VLA data efficiency and generalization.
ARFM [141]	2025	Adaptively adjusts RL signal weights, stabilizing offline RL fine-tuning for flow-based VLAs.
VLA-RL [142]	2025	Improves autoregressive VLAs via multimodal trajectory modeling and online RL.
RPD [151]	2025	Uses a VLA teacher to guide an RL student policy, achieving policy distillation and performance gains.
SRPO [143]	2025	Uses successful in-batch trajectories as self-reference, constructing process rewards for failed trajectories.
LongNav-R1 [152]	2026	Models navigation as multi-turn RL and uses HAPO to improve long-horizon credit assignment.
World Model		
EA-WM [159]	2026	Maps actions and motion states into structured visual representations for better spatial modeling.
Hi-WM [161]	2026	Uses world-model rollbacks and corrective trajectories for policy post-training.
WM-Dagger [162]	2026	Generates OOD recovery data with correction and consistency filtering.
H-WM [158]	2026	Jointly predicts logical transitions and visual subgoals for long-horizon tasks.
ContactGaussian-WM [156]	2026	Learns physically constrained contact dynamics from sparse video using Gaussian representations.
DDP-WM [155]	2026	Separates key dynamics from background updates for efficient planning.
OSVI-WM [160]	2025	Predicts future latent states from one demonstration and decodes them into waypoints.
LaDi-WM [154]	2025	Predicts future latent states and iteratively improves manipulation policies.
PIN-WM [157]	2025	Estimates rigid-body dynamics parameters via differentiable physics to enhance Sim2Real policy transfer.
WoVR [163]	2026	Uses the world model as an RL environment while controlling imagination errors.
Long-Horizon		
Anticipation-VLA [165]	2026	Guides VLAs through long-horizon execution via adaptive recursive subgoal generation.
LoHo-Manip [166]	2026	Uses receding-horizon subtask and trajectory prediction for implicit progress tracking.
HELM [168]	2026	Combines episodic memory, pre-execution checks, and rollback replanning to reduce long-horizon errors.
Goal2Skill [174]	2026	Links high-level VLM planning with low-level VLA skills in a reflective closed loop.
RoboClaw [173]	2026	Unifies data collection, policy learning, and skill scheduling to support long-horizon tasks.
VQ-Memory [169]	2026	Compresses proprioceptive history into discrete memory tokens.
TempoFit [170]	2026	Reuses the hierarchical KV cache of a frozen VLA to achieve training-free temporal memory augmentation.
Critic in the Loop [172]	2026	Uses a visual Critic for planning, anomaly detection, and replanning.
KC-VLA [171]	2026	Uses key historical frames to model long-horizon dependencies.
LiLo-VLA [167]	2026	Separates global planning from local interaction and improves generalization through skill reuse.
AtomVLA [131]	2026	Combines atomic subtask decomposition with world-model scoring for long-horizon execution.
BagelVLA [175]	2026	Interleaves language planning, visual prediction, and action generation for multi-stage tasks.

memory augmentation have provided some mitigation, most still rely on external offline modules and have not yet formed a stably integrated closed-loop execution mechanism with the end-to-end backbone.

C. Bottlenecks in Lightweight Deployment and Real-Time Control

VLAs are generally built on large-scale vision-language models; while this improves semantic understanding, it also brings high inference latency, making it difficult to meet the real-time requirements and edge-deployment constraints of closed-loop robotic control. Although specialized compression methods for VLAs have begun to emerge, achieving extreme compression while maintaining action-control precision remains an urgent engineering challenge to be solved. Relatively little work has specifically designed control frequency and closed-loop stability for action prediction; meanwhile, the "slow high-level planning, fast low-level execution" hierarchical architecture is not yet mature, task-adaptive computation and asynchronous inference mechanisms remain at an exploratory stage,

and a systematic solution to balancing performance and efficiency is still lacking.

D. Lack of Continual Learning Ability under the Static-Data Paradigm

Most current VLAs still adopt an offline training pipeline based on static datasets, with the model deployed directly after training and lacking mechanisms for continuously recycling and utilizing new tasks, failure cases, and recovery experience encountered during execution. As a result, models find it difficult to continuously improve from errors made in actual deployment, and equally difficult to promptly adapt to new scenes and new failure modes that constantly emerge in open environments. Although some research has explored online interaction, failure-trajectory generation, and policy retraining, establishing a stable closed-loop data pipeline of "deploy first, then gather feedback, then filter, and finally train" remains an important problem for VLA continual learning and scaled application.

E. Insufficient Modeling of Physical Laws

Existing VLAs mainly rely on data-driven statistical learning and lack explicit modeling of physical constraints such as contact dynamics, support relations, friction properties, and mass distribution. Many manipulation failures do not stem from errors in perception or semantic understanding, but from the model's lack of physical-level prediction of action consequences—a problem that is especially prominent in fine-grained tasks such as grasp stability, assembly alignment, and deformable-object manipulation. How to effectively incorporate physical priors and affordance knowledge into representation learning remains at a relatively early stage of exploration, constraining the stability and interpretability of models in contact-rich scenarios.

F. Lack of Embodied Agentic Capability

Most current VLAs still center on a policy mapping from vision-language input to robot action, with agentic capabilities such as task decomposition, long-term memory, tool invocation, and active exploration typically implemented via additional modules. In complex tasks involving multi-stage planning and human-robot collaboration, a stable, unified collaborative architecture among high-level reasoning models, low-level VLA policies, and world models has not yet formed; the interface between high- and low-level information, cross-module memory sharing, and safety-constraint mechanisms remain incomplete. These issues limit the development of robots from executing a single policy toward becoming autonomous closed-loop systems with planning, verification, and recovery capability.

G. Scarcity of Real-World Data and Insufficient Credibility of Alternative Data Sources

Robotic manipulation data depends on real platforms and expert demonstrations, making collection costly and difficult to scale; differences among embodiments in kinematic structure and action space also make data difficult to reuse across platforms, and coverage is especially scarce for tasks such as complex contact manipulation and deformable-object handling. Although human video and simulation data can be scaled at low cost, the former faces a cross-embodiment mapping challenge due to differences between human and robot kinematic structure, while the latter is constrained by the sim-to-real gap in both visual appearance and physical dynamics; more fundamentally, both data sources lack a stable supervisory signal from visual dynamics to executable robot action, constraining their direct and effective use.

V. Application Scenarios for VLAs

The evolution of VLA capabilities must ultimately be tested through deployment in the real physical world. Different application scenarios impose different core requirements on robots: household service emphasizes continuous

adaptation in an open world; surgical robots emphasize predictable behavior under extreme precision constraints; autonomous driving emphasizes visual grounding and anticipatory decision-making; and industrial manufacturing emphasizes reliability and flexibility under large-scale deployment. This chapter focuses on these four representative scenarios, illustrating how the technical evolution of VLAs responds to the real-world constraints of different domains.

A. Household and Service Robots

The core challenge of the household environment lies in its openness: object types are diverse, spatial layouts vary, and task requirements are highly individualized, requiring the robot to accomplish spatial understanding, instruction execution, and long-term autonomous adaptation without structured priors. This requires VLAs to possess not only fine-grained local manipulation ability, but also stable performance in unseen environments, along with the ability to continuously improve as deployment time progresses. Around this need, recent research unfolds along three levels: at the level of spatial representation, SG-VLA [176] enhances the robot's global-local perception of household scenes by fusing multi-view observations with auxiliary supervisory signals; at the level of open-world generalization, $\pi_{0.5}$ [177] enables the model to transfer knowledge from other environments to unseen household layouts and objects through joint training on heterogeneous, cross-scene data; at the level of long-term deployment, $\pi_{0.6}^*$ [178] continuously integrates online interaction experience and expert corrective feedback, driving robots to improve execution efficiency on complex, long-horizon tasks through real-world use. These works show that household service scenarios are pushing VLAs from a "train once, deploy statically" paradigm toward "continually learning, lifelong-adapting" embodied intelligence systems.

B. Surgical and Medical Assistance Robots

Surgical and medical assistance scenarios impose requirements on precision, safety, and data availability that exceed those of most application domains: operational error must be controlled at the sub-millimeter scale, behavior must be highly predictable, and the privacy sensitivity of patient data limits the scale of available training data. This combination of constraints determines that the technical route for medical VLAs differs from general-purpose models that pursue broad generalization, tending instead toward specialized structural design and data-efficient learning. In this direction, researchers have used lightweight, specialized network structures to demonstrate that models can achieve stable, fine-grained manipulation with very little demonstration data [179]; Cosmos-H-Surgical [180] constructs a surgery-specific world model combined with inverse-dynamics inference, converting large volumes of unlabeled surgical video into usable training signal, alleviating the scarcity of action-labeled

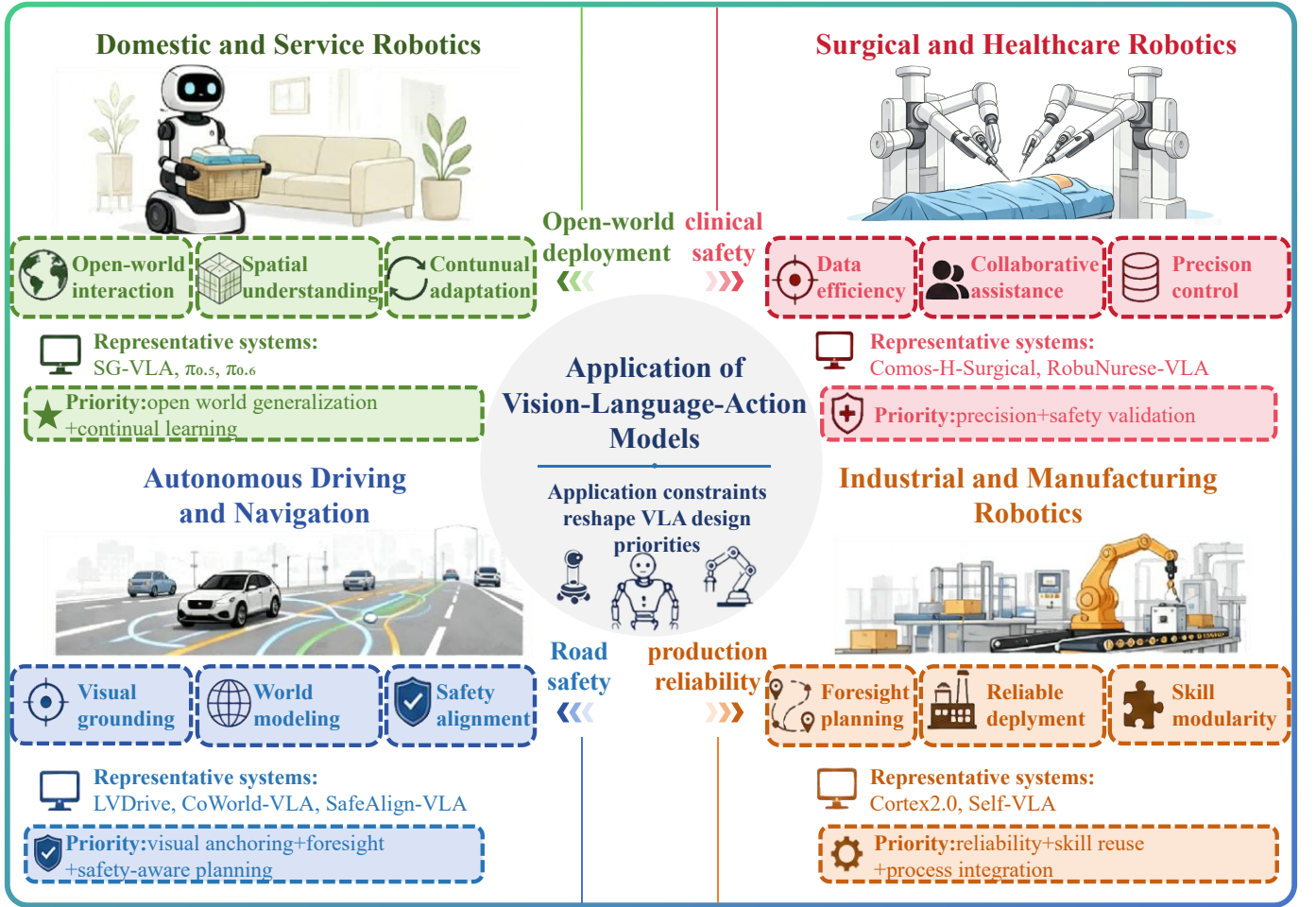


Fig. 5. Application landscape and design priorities of VLAs across four representative domains. A systematic overview of VLA model deployment topologies across domestic service, surgical operations, autonomous driving, and industrial manufacturing. The diagram illustrates how practical domain requirements—such as open-environment adaptation, clinical safety, road safety, and production reliability—determine the design priorities of VLA models, directing them toward generalization, control precision, foresight planning, and skill reuse, respectively, alongside representative systems in each field.

data; RoboNurse-VLA [181] integrates visual segmentation with language understanding to achieve surgical-instrument recognition and real-time delivery based on natural spoken instructions, providing a new interaction mode for human-robot collaboration in the operating room. The development path of medical VLAs reflects a domain characteristic of being “specialized rather than general, precise rather than broad.”

C. Autonomous Driving and Navigation

Autonomous driving requires a model to complete scene understanding, risk assessment, and trajectory decision-making in real time under high-dimensional perceptual input, where errors can lead to irreversible safety consequences; this domain therefore imposes comprehensive requirements on the use of visual information, long-horizon scene prediction, and risk-sensitive decision-making mechanisms. Around these requirements, research on driving VLAs has formed several technical threads: some work aims to correct the model’s over-reliance on language and vehicle state, redesigning the form of action supervision

and the trajectory-decoding interface to encourage the model to make fuller use of visual information for planning [182], [183]; other work introduces world-modeling capability, enhancing the model’s anticipatory ability for complex traffic scenes through multi-expert scene representations or generative future-frame prediction, alleviating the lack of foresight in reactive policies [184], [185]; on the safety-enhancement side, related work explicitly introduces risk identification, behavior correction, and external safety priors into the planning process through counterfactual negative-sample construction, a trajectory generate-evaluate-optimize mechanism, and retrieval of historical driving experience, thereby improving decision safety and robustness of driving VLAs in complex scenarios and long-tail conditions [186]–[188]; in addition, a batch of work explores unified architectures for language reasoning and continuous control, using low-entropy meta-actions, single-forward-pass joint generation, or mixture-of-experts parameter decoupling to balance performance with inference efficiency and multi-task capability [189]–[191]. Overall, the research focus of driving VLAs is

shifting from direct adaptation of general-purpose vision-language models toward specialized architecture design oriented toward driving perception, reasoning, and control requirements.

D. Industrial and Manufacturing Robots

The core requirement that industrial manufacturing scenarios place on VLAs is reliability and flexibility under conditions of large-scale deployment: robots must maintain stable operation in unstructured environments with frequent clutter and occlusion and dense contact, while also adapting to constantly changing task requirements at low configuration cost. Around these goals, existing research mainly unfolds along two complementary paths. The first line of work focuses on improving execution reliability in complex tasks, generating and evaluating candidate trajectories in a visual latent space and selecting subsequent actions based on expected success rate and execution efficiency, thereby mitigating error accumulation and failure propagation in long-horizon tasks under reactive policies; at the same time, work oriented toward real production lines forms a deployment-driven iterative optimization loop through continuous on-site data collection, sample screening, policy fine-tuning, performance evaluation, and recovery-data supplementation, pushing industrial VLAs from one-off training followed by direct execution toward a deployment mode combining anticipatory planning with continuous optimization [192], [193]. The second line of work focuses on enhancing task configuration and skill-reuse capability, decomposing complex assembly or disassembly processes into callable, specialized skills through structured skill libraries and language-driven task scheduling, dynamically switching based on task context, thereby improving the robot's adaptability to high-precision, heavily contact-rich, and multi-stage operations [194], [195]. Overall, industrial VLAs are not blindly pursuing a fully end-to-end unified policy, but are gradually adopting a compositional architecture in which a foundation model handles task understanding and planning while specialized skill modules perform precise operations. This development direction better matches the long-term engineering demands of industrial systems for reliability, maintainability, and task flexibility.

VI. Future Outlook

Although VLAs have made clear progress in language-conditioned control, multi-task learning, and cross-scene transfer, there remains a considerable gap before reaching a general-purpose embodied intelligence system with broad generalization ability, stable deployment ability, and continual learning ability. Most existing methods rely on large-scale vision-language models and robotic action data, achieving perception, instruction understanding, and action generation through end-to-end training. Faced with long-horizon tasks, continuous interaction, and dynamic change in real environments, predicting actions directly from the current observation alone often fails to maintain

stable decision-making and is not conducive to accumulating experience or updating capability. Therefore, future research should not focus solely on model scale, but should simultaneously improve architecture design, data construction, physical modeling, deployment methods, and agentic collaboration mechanisms. Based on the issues discussed above, the following sections discuss potential development directions for VLAs along seven dimensions.

A. Generalization: From In-Distribution Success Toward Robustness in Open Environments

Generalization ability is the core prerequisite for VLAs to become general-purpose robots. Existing models perform well within the training distribution, but performance still drops markedly when facing unseen tasks, scenes, robot embodiments, and long-tail perturbations. Future research therefore needs to focus on cross-task, cross-scene, cross-embodiment, and long-tail robustness. Cross-task generalization requires models to learn reusable skills and compose them; cross-scene generalization requires models to adapt to different environments such as households and factories; cross-embodiment generalization requires models to adapt to different robot platforms. Long-tail robustness concerns real-world scenarios such as occlusion, abnormal layout, human interference, and failure recovery. Future work needs to develop unified action representations and transferable control interfaces, build a data ecosystem that includes both success and failure cases, and design more challenging compositional-perturbation evaluations. In addition, by modeling environmental dynamics and action consequences, WAM is expected to improve a model's prediction and transfer ability in unseen environments. Ultimately, the goal of VLAs will shift from high success rates on fixed benchmarks toward stable, reliable behavior in open environments.

B. Lightweight Deployment: From Large-Scale Models Toward Real-Time, Deployable Embodied Policies

Current VLAs often rely on large-scale vision-language models; while this improves semantic understanding, it also brings high computational cost and inference latency. Robotic control must meet real-time and edge-deployment requirements, making lightweight design a key direction. Methods such as low-rank adaptation, quantization, pruning, and knowledge distillation can reduce training and deployment cost. At the same time, mechanisms such as action-chunk prediction, asynchronous inference, and streaming control also help improve closed-loop control efficiency. Future VLAs may form a hierarchical architecture of "large model thinks slowly, small model executes quickly": a high-level model handles planning and reasoning, while a low-level controller handles real-time execution. Through task-adaptive computation, VLAs are expected to achieve a balance between performance and efficiency.

C. Data Flywheel: From Static Datasets Toward a Continuously Evolving Data Engine

Existing VLAs mainly rely on static datasets for training, but the real environment continuously produces new tasks and failure cases, and fixed data cannot support continual evolution. Future work therefore needs to build a data flywheel to enable continual learning. A data flywheel updates the model by continuously recycling deployment experience: as a robot executes tasks, it records success and failure cases, which are filtered and used in the next round of training, after which the improved policy is redeployed in practice. In the future, competition among VLAs will be not only competition among models, but also competition in data infrastructure. Combined with WAM and world models, it becomes possible to predict failure risk and filter out implausible trajectories, thereby improving data-utilization efficiency and safety, evolving VLAs from one-off trained models into continuously evolving embodied learning systems.

D. Physical Knowledge Enhancement: From Data-Driven Toward Modeling Physical Laws

Current VLAs mainly rely on data-driven learning, but the real world contains complex contact relations and dynamic constraints, and relying solely on statistical regularities makes it difficult to fully understand the physical environment. Many robot failures do not stem from perceptual error but from a lack of understanding of physical laws. Future VLAs need to incorporate physical knowledge so that models can understand support relations, friction properties, mass distribution, and contact dynamics. For example, in grasping, assembly, and non-rigid-body manipulation, robots need to predict the consequences of actions and comply with physical constraints. Research directions include incorporating physical priors and affordance knowledge into representation learning, combining this with world models for future-state prediction, and using physical simulation to generate high-quality training data. Through the joint drive of data and physics, VLAs are expected to gain greater stability and interpretability.

E. Long-Horizon Planning: From Short-Horizon Task Execution Toward Complex Task Reasoning

Current VLAs perform well on short-horizon manipulation tasks, but real-world tasks often involve multi-stage decision-making and long-term goals. Household service and industrial assembly tasks, for example, both require task decomposition, state memory, and error-recovery ability. Future VLAs need to build hierarchical policies: a high level handles task planning, a low level handles action execution, and a world model is used to predict future states and evaluate the consequences of actions. Long-horizon tasks still face issues such as error accumulation and state forgetting, so long-term memory, closed-loop replanning, and post-execution reflection mechanisms need to be introduced. At the same time, evaluation standards should also expand from a single success rate to process

quality, recovery ability, and long-term consistency, for a more comprehensive assessment of model capability.

F. Agentification: From VLA Policy Models Toward Autonomous Embodied Agents

Future VLAs will evolve from single policy models into autonomous embodied agents. A truly capable agent should not only execute instructions, but also possess task decomposition, long-term memory, tool invocation, active exploration, and human-robot collaboration ability. Under this framework, the VLA is responsible for action execution, an LLM or VLM is responsible for high-level reasoning, a WAM or world model is responsible for prediction and verification, and a memory module maintains historical experience and task state. VLAs will become an important execution component of embodied agents. However, agentification still faces challenges such as the interface between high and low levels, memory management, safety constraints, and human-robot interaction. Future research needs to further refine system architecture and safety mechanisms, pushing robots toward autonomous closed-loop systems.

G. Synthetic Data: From Low-Cost Scaling Toward Physically Credible Data Generation

Robot data collection is costly, especially for complex manipulation and hazardous scenarios, making synthetic data an important avenue for scaling VLA capability. Through simulated environments, procedural generation, and generative models, large-scale training data can be obtained at low cost. The value of synthetic data lies not only in scale, but also in supplementing long-tail scenarios that are underrepresented in real data, such as abnormal layouts, failure recovery, and hazardous tasks. It can also support tasks such as cross-embodiment transfer and complex contact manipulation. However, discrepancies between simulation and reality may cause a model to learn incorrect regularities. Future research therefore needs to focus on data quality, physical credibility, and real-world transferability, calibrating through physical verification and real feedback. Furthermore, synthetic data should be combined with the data flywheel and world models, generating targeted samples from real-world deployment failure cases to achieve "real-feedback-driven data generation." This will help overcome the limitation of scarce real-world data and improve model generalization.

Collectively, the future development of VLAs will shift from optimizing a single model toward the co-evolution of models, data, and agentic systems. Generalization ability determines the scope of adaptation to open environments; lightweight design determines deployment efficiency; the data flywheel determines continual-learning ability; physical-knowledge enhancement determines the depth of understanding of environmental dynamics; long-horizon planning determines the ability to execute complex tasks; agentification determines autonomy; and synthetic data determines the boundary of capability expansion. Along this direction, VLAs are expected to gradually

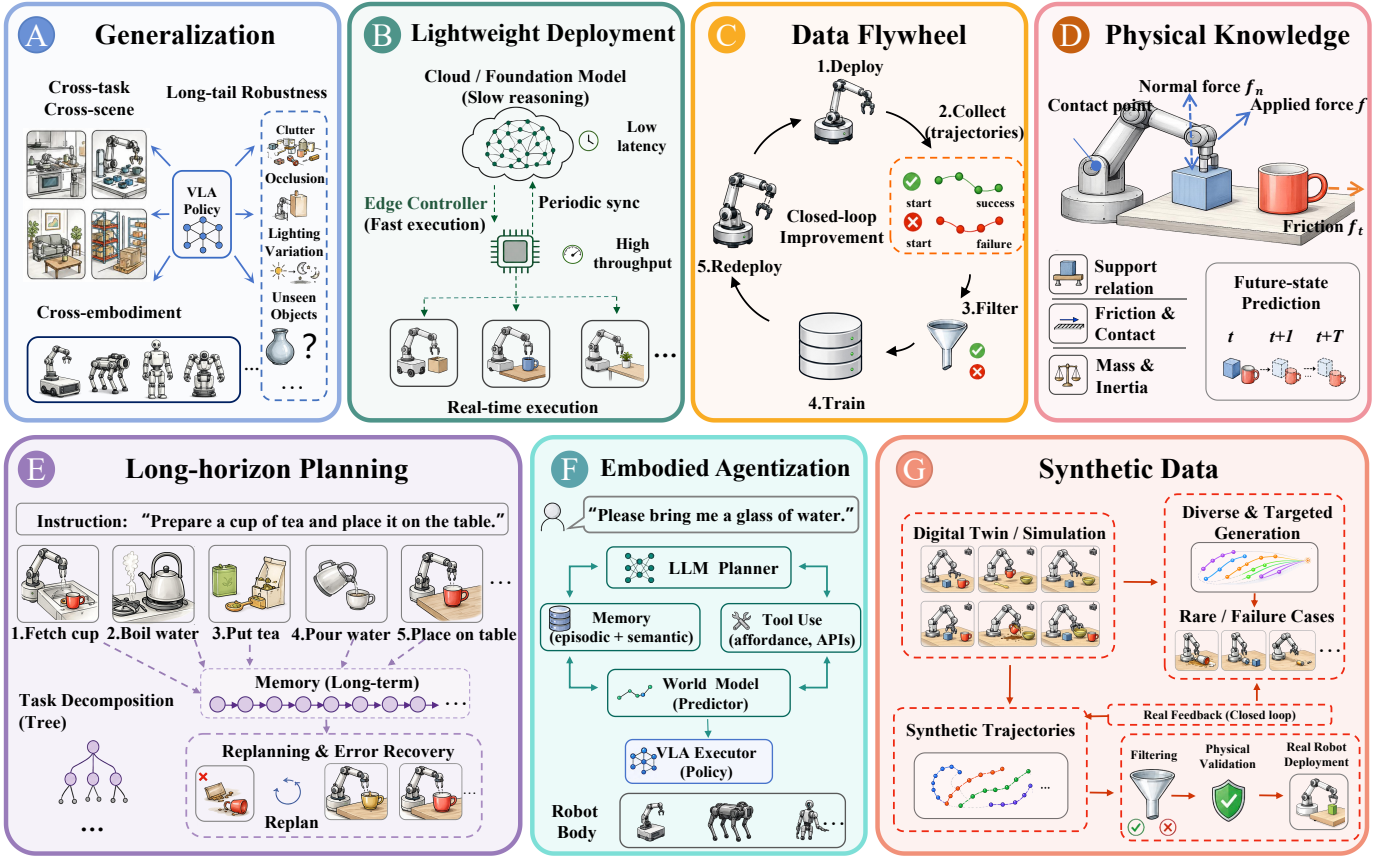


Fig. 6. Open research directions and critical challenges in VLAs. A systematic summary of seven foundational pathways enabling next-generation generalist embodied models: open-world generalization, lightweight deployment and real-time control, continuous data flywheels, physical knowledge enhancement, long-horizon planning and memory, autonomous embodied agentization, and physically faithful synthetic data generation.

develop into next-generation embodied foundation models with predictive, generalizable, and continually learning capabilities.

VII. Conclusion

Our survey indicates that model scale, data coverage, and benchmark success rates are each insufficient, when considered in isolation, to characterize the overall capability of VLAs in real-world environments. Large-scale vision-language pretraining improves robots' understanding of task semantics and facilitates knowledge transfer, yet these advantages do not necessarily translate into stable control in open-world settings. No single action-decoding paradigm is universally superior; rather, its suitability depends on the task-specific requirements for semantic modeling, action accuracy, and inference efficiency. Although model lightweighting and data scaling alleviate computational and data constraints, respectively, neither is sufficient to ensure both control stability and robust generalization in real-world environments.

Accordingly, practical deployment requires not only the generation of executable actions, but also stable closed-loop control under limited computational resources and consistency between the generated actions, the current environmental state, and the task constraints. The development of VAM and WAM reflects an important shift

in robot policy learning, whereby video dynamics and future states are increasingly incorporated into the action-generation process. Their significance lies not merely in appending a video predictor or world model, but in using representations of future states to constrain action generation and to provide intermediate information that can be evaluated during predictive planning. Future research should therefore advance model design, data feedback loops, and evaluation methodologies in a coordinated manner. Data pipelines should continuously incorporate deployment failures and recovery experiences; evaluation protocols should jointly consider prediction quality, control efficiency, and task success; and model design should mitigate the propagation of world-prediction errors into action policies. By strengthening the interaction among semantic understanding, action generation, and environment-dynamics modeling, while introducing continual updates driven by real-world feedback, VLAs may evolve into embodied agents capable of stable decision-making and sustained adaptation in open environments.

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